

BENZENE — THE SIX-CARBON RING BEHIND MODERN CHEMISTRY

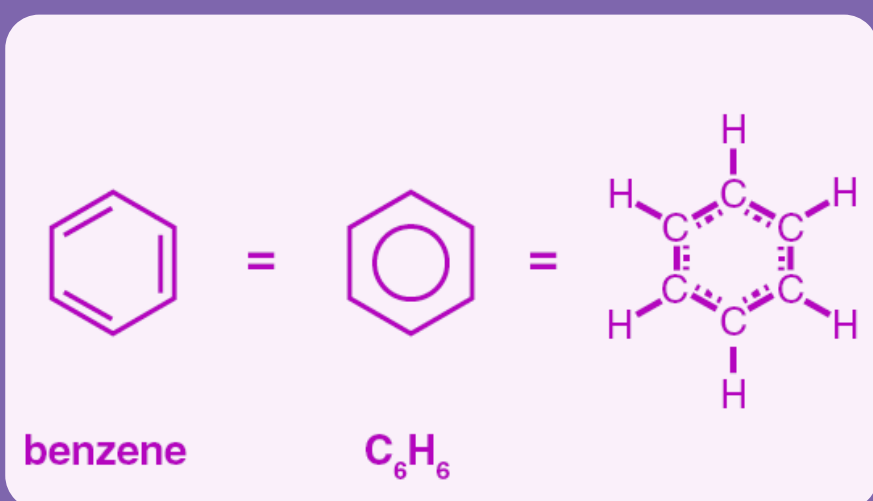
COMPOUND IDENTITY

Common name: Benzene
IUPAC name: Benzene
Molecular formula: C_6H_6
Molar mass: 78.11 g/mol
CAS number: 71-43-2

IMPORTANT PROPERTIES

- Colorless liquid at room temperature
- Aromatic hydrocarbon
- Nonpolar
- Slightly soluble in water
- Highly flammable
- Boiling point $\approx 80.1^\circ C$

STRUCTURE



Benzene contains six carbon atoms arranged in a planar ring with delocalized π electrons, giving it aromatic stability.

USES & VALUE

Benzene is mainly important as an industrial chemical feedstock.

Examples include production of chemicals used to make:

- Plastics
- Resins
- Synthetic fibers
- Detergents
- Dyes
- Other industrial chemicals

WHAT IS AMAZING ABOUT IT?

Its six π electrons are delocalized around the ring.

Instead of three isolated double bonds, the electrons are shared across the ring, giving benzene extra stability and unique reactivity



RISKS & SAFETY



BENZENE IS A KNOWN HEALTH HAZARD

- Highly flammable
- Harmful through exposure
- Long-term exposure can damage blood-forming tissues
- Known human carcinogen

CHEMICAL ENGINEERING CONNECTION

Petrochemical Processing & Separation

Chemical engineers may encounter benzene in:

PETROLEUM FEED ➔ **SEPARATION** ➔

BENZENE-CONTAINING STREAM



PETROLEUM FEED

- Petrochemical processing
- Distillation/separation
- Chemical manufacturing
- Process safety
- Waste treatment

“A SIMPLE SIX-CARBON RING BECAME ONE OF THE MOST IMPORTANT BUILDING BLOCKS OF MODERN CHEMICAL MANUFACTURING.”

REFERENCES

PubChem. (n.d.-c). Benzene. PubChem. <https://pubchem.ncbi.nlm.nih.gov/compound/71-43-2>

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